



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI MURUGAN AUTO FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

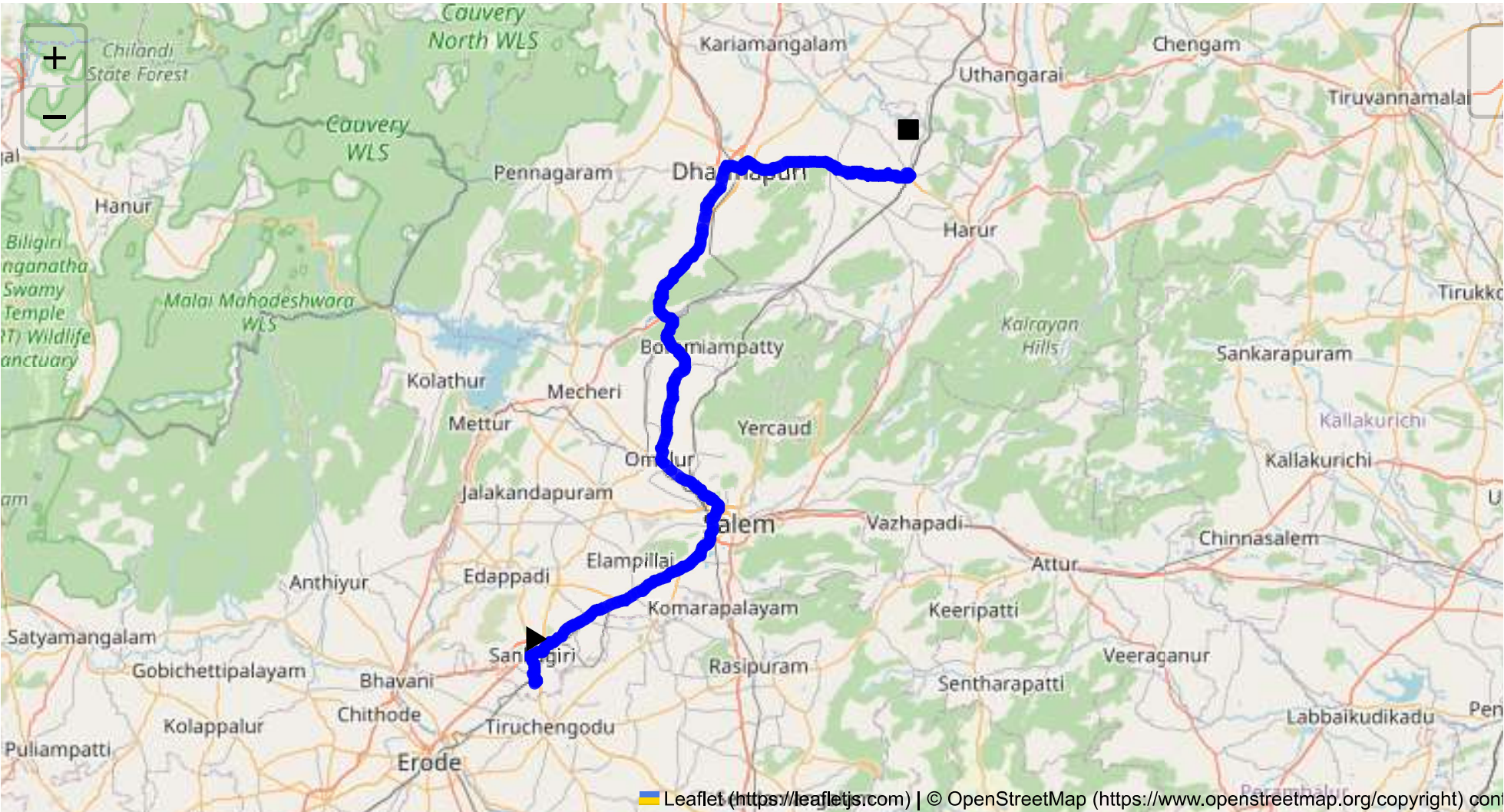
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 136.18 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.131822, 78.390915)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Kosapatty in Tamil Nadu covers approximately 136.18 kilometers and takes about 2.69 hours under normal conditions for heavy vehicles carrying hazardous materials. The journey progresses through NH544 and various state highways, primarily in a rural setting, interspersed with small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with the following key weather patterns:

- **Dry Season (January to April):** Minimal rainfall, favorable for transportation.
- **Hot Season (May to August):** High temperatures can impact road conditions and driver performance.
- **Monsoon Season (September to December):** Heavy rains can cause waterlogging, slippery roads, and potential landslides, especially in hilly or poorly drained areas.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM).
- **Congestion Areas:** Potential bottlenecks include the urban centers you pass, such as Erode and smaller towns with narrow roads.
- **Avoidance Tips:** Plan travel outside peak hours or use bypasses where available.

4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally well-maintained, but varying quality across state roads.
- **Bridges and Overpasses:** Inspect for weight limitations and structural integrity.
- **Potholes and Repairs:** Frequent in rural road sectors; maintenance inconsistency can affect travel speed and safety.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route 1:** Deviate through NH44 for a more direct route in severe weather or road blockage scenarios.
- **Alternative Route 2:** Local state roads connecting through inland paths provide detours avoiding major towns.

6. Local Regulations Affecting Hazardous Material Transport

- Permits required for transporting hazardous materials.
- Restrictions likely in residential and densely populated areas.
- Scheduled night travel might be mandated to lower risk.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

The region has a moderate history of incidents involving heavy vehicles due to rain-induced road damage or vehicle overturns during sharp turns in rural segments. Fatal incidents involving hazardous materials are rare but possible under poor weather or mechanical failures.

8. Environmental Considerations and Sensitive Areas

- **Protected Wildlife Zones:** Be aware of restrictions around reserved forests.
- **Agricultural Areas:** Watch for local farming traffic and livestock crossings.
- **Water Bodies:** Respect no-pollution zones near rivers and lakes, particularly during heavy rains.

9. Analysis of Communication Coverage

- **Good Coverage:** Along national highways and urban centers.
- **Potential Dead Zones:** Rural stretches, particularly between villages without cell towers.

10. Estimated Emergency Response Times

- **Urban Areas:** Erode or large towns - approximately 30-45 minutes.
- **Rural Areas:** Expect delays up to 1-2 hours depending on road conditions and remoteness.

12. Overall Risk Assessment Summary

The route presents moderate risks due to frequent weather changes, varying road quality, and potential traffic congestion in towns. However, with proper precautions, realistic timing, and adherence to local transport regulations, these risks can be mitigated. Notable care should be taken during monsoon seasons and on rural stretches where emergency services may have delayed reachability. Always ensure communication means remain operative and map alternate routes for any foreseeable obstructions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.53501, 77.95707	30 KM/Hr
12	Turn	Medium	11.53601, 77.95856	30 KM/Hr
13	Turn	Medium	11.62403, 78.10893	30 KM/Hr
14	Turn	High	11.64180, 78.11808	15 KM/Hr
15	Turn	High	11.64325, 78.11993	15 KM/Hr
16	Turn	Medium	11.69519, 78.10775	30 KM/Hr
17	Turn	Medium	11.73958, 78.05105	30 KM/Hr
18	Turn	Medium	11.75835, 78.05016	30 KM/Hr
19	Turn	Medium	11.86847, 78.08367	30 KM/Hr
20	Turn	Medium	11.87942, 78.08197	30 KM/Hr
21	Turn	Medium	11.92585, 78.06507	30 KM/Hr
22	Turn	Medium	11.93704, 78.05402	30 KM/Hr
23	Turn	High	11.95818, 78.04978	15 KM/Hr
24	Turn	Medium	11.95988, 78.04948	30 KM/Hr
25	Turn	Medium	11.96304, 78.05136	30 KM/Hr
26	Turn	High	11.96399, 78.05135	15 KM/Hr
27	Turn	High	11.96580, 78.04923	15 KM/Hr
28	Turn	Medium	11.96675, 78.04939	30 KM/Hr
29	Turn	Medium	11.96791, 78.05070	30 KM/Hr
30	Turn	Medium	12.14026, 78.13955	30 KM/Hr
31	Turn	Medium	12.14048, 78.13998	30 KM/Hr
32	Turn	High	12.14499, 78.16500	15 KM/Hr
33	Turn	Medium	12.14721, 78.26926	30 KM/Hr
34	Turn	Medium	12.14553, 78.27731	30 KM/Hr
35	Turn	Medium	12.14597, 78.27877	30 KM/Hr
36	Turn	Medium	12.14467, 78.28372	30 KM/Hr
37	Turn	High	12.13069, 78.31362	15 KM/Hr
38	Turn	Medium	12.13248, 78.31918	30 KM/Hr
39	Turn	Medium	12.12880, 78.33678	30 KM/Hr
40	Turn	Medium	12.12942, 78.34092	30 KM/Hr
41	Turn	High	12.12751, 78.35960	15 KM/Hr
42	Turn	Medium	12.12857, 78.36048	30 KM/Hr
43	Turn	Medium	12.12645, 78.39367	30 KM/Hr
44	Turn	High	12.12746, 78.39480	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
38	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
39	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
40	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
41	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
42	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
43	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
44	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
45	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
46	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
47	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
48	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
49	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
50	school	Marudham Matric Hr Sec School	12.1251245, 78.3823077	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.13658, 78.16171



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14589, 78.17156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62403, 78.10893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92585, 78.06507



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96399, 78.05135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96675, 78.04939



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96791, 78.05070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14026, 78.13955



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14048, 78.13998



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14499, 78.16500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14721, 78.26926



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14553, 78.27731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14597, 78.27877



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14467, 78.28372



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13069, 78.31362



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13248, 78.31918



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12880, 78.33678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12942, 78.34092



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.12751, 78.35960



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12857, 78.36048



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12645, 78.39367



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.12746, 78.39480